Name: _____

Exercise 7: Change Detection

Introduction

In this Exercise we will look at the problem of change detection using several common analytical techniques. Some of these techniques will look familiar from earlier exercises; others are new this week.

The subject of analysis is Yellowstone National Park. You'll be working with Landsat TM images from the autumn of 1987 and the autumn of 1988. The Park experienced a transformative event between these two images, and we will use Landsat images, combined with a digital elevation model and some ancillary GIS data, to detect, quantify, and understand this event.

1. Exploring with the 3D viewer

As in the previous labs, download the Exercise 7 zip file from Canvas and extract all files to a local directory.

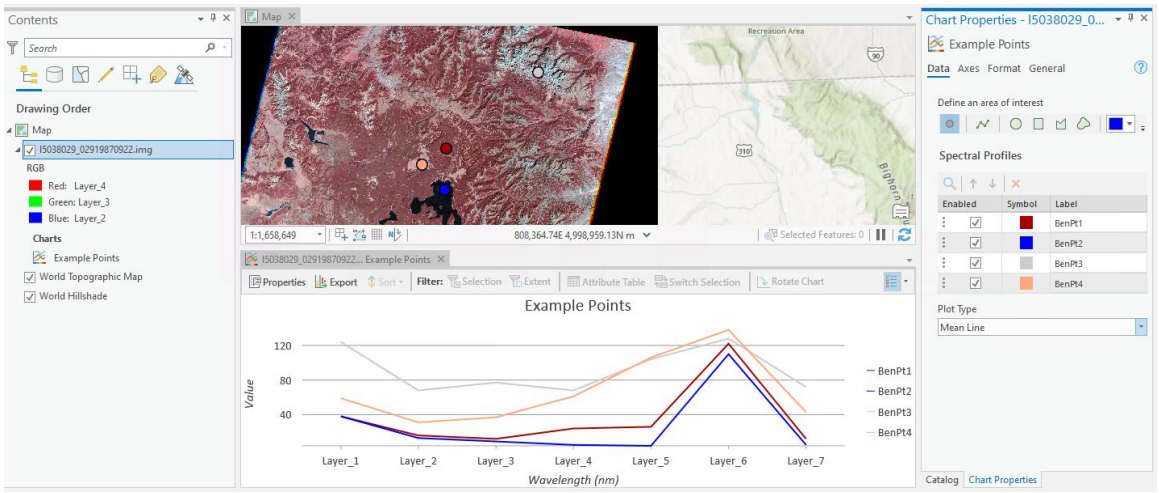
Start ArcGIS Pro, create a New Map, and use Add Data to display the 1987 Landsat image: L5038029_02919870922.img. Change the Band Combination (under *Appearance*) to a custom RGB-432 scheme.

Q1: What are the projection and spatial resolution of the image? What are the approximate latitude / longitude at the center of the image? (projection and resolution details are available by double-clicking the file name in Contents and looking under Source, and lat/lon can be viewed by changing the units of the scroll coordinates at the bottom of the project window)

Explore the spectral character of land covers in the scene. Right-click on the file name in *Contents*, selecting *"Create Chart"* → *"Spectral Profile"* and then choosing an area of interest in the *Chart Properties* dialog. The area of interest can be a point, line, or shape. You can create several Areas of interest and then adjust their name and color in the *Chart Properties* dialog. You can type in any Labels that you want, but note that you need to hit the Tab key after typing in the name to get it to stick.

This is a Landsat 5 TM image, so bands 1-7 are Blue, Green, Red, Near IR 1, Near IR 2, Thermal IR, Mid IR.

Q2: List some of the major land cover types present in the scene, along with one key reflectance characteristic that might be useful in identification / classification. Include the spectral profile chart in your answer.

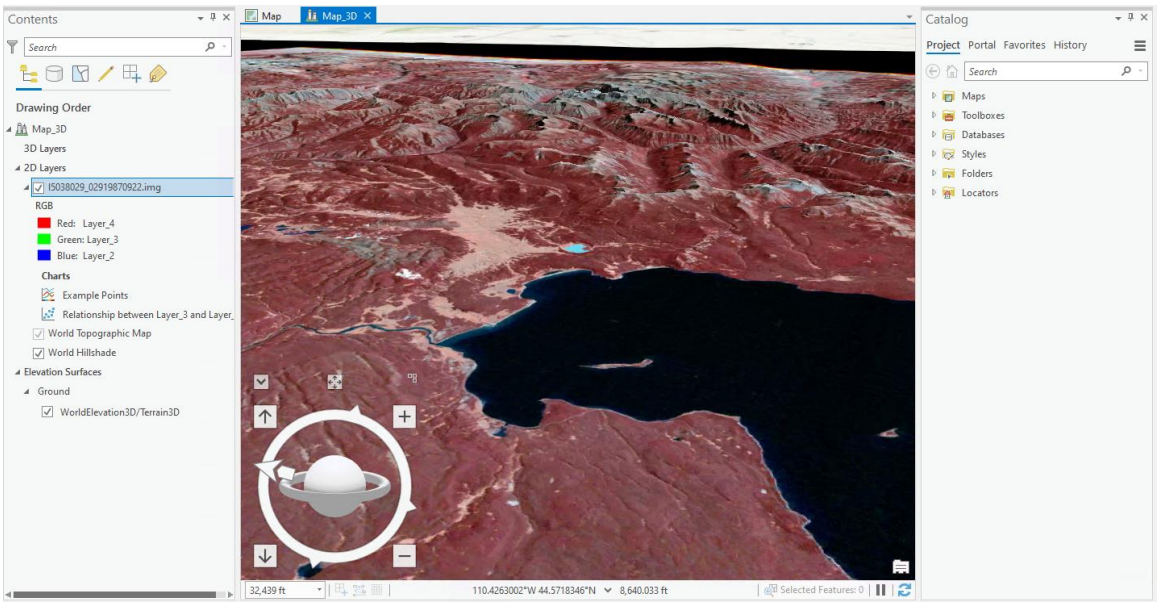


The Yellowstone area is characterized by active geological conditions and dramatic topography. We won't get into details of geological analysis in this exercise, but it can still be useful to explore your image in the context of topographic relief. We'll do this using the ArcGIS Pro "image drape" interface:

- Under the *View* menu, select *Convert → To Local Scene.*
- A new navigation compass will appear in the bottom left of your Map_3D window. Click on the "Show Full Control" icon (the up arrow in the image below)



- This will give you a new inner circle on the compass that can be used to adjust your view angle and direction. The up and down arrows control your elevation and the plus and minus icons control your zoom.
- Explore your image in 3D. Note that ArcGIS is using a default elevation dataset. We could drape over our own custom elevation dataset if we needed more detail, but this is sufficient for our purposes.



Q3: In one or two sentences, describe the general topographic setting of the scene.

Q4: How did the image drape help your answer to Q3 (if at all)? Could you have provided the same description on the basis of the Landsat image alone?

Return to your standard "Map" view for the following work.

2. Clip to an area of interest

The Landsat scene that you are working with captures an area larger than Yellowstone NP. For the rest of our analyses, we want to focus on the area within the park's borders. To do this, we'll need to bring in ancillary information that will allow us to constrain our analysis to the portion of the image that falls within the park.

- Use *Add Data* to add yellowstone.shp to your map. This provides the boundary of the park.
- Click on *Tools* under the *Analysis* menu.
- Under *Toolboxes* in the *Geoprocessing* dialog, select *Data Management Tools → Raster → Raster Processing → Clip Raster*
- Use the Landsat image as your *Input Raster* and the *yellowstone* shapefile for *Output Extent*. Select "Use Input Features for Clipping Geometry."
- Select an output file name and run the tool.
- Your clipped image will appear in the *Contents*. Change it to an RGB-432 band combination, and turn off the *yellowstone* shapefile and the original image.

Now, add the image 1988clip_park.img to your project.

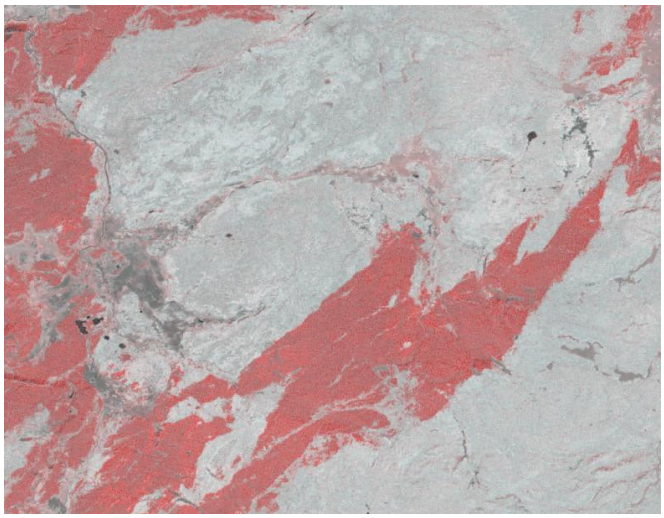
Q5: Are there any significant differences between the 1987 and the 1988 images?

3. Change Detection

We will now attempt to quantify differences between the 1987 and 1988 images using two different approaches.

NDVI Change

- Under the *Imagery* menu, select *Raster Functions* and chose *NDVI*.
- Pick the visible and infrared input bands that are appropriate for Landsat TM. Choose "Scientific Output" so that you get true NDVI values and not just a pretty looking stretch.
- Calculate NDVI for each of your clipped images. You can "create new layer" or "Save as." It doesn't matter for this exercise.
- Calculate a difference: select your two NDVI layers (Control+click) in the *Contents*, then select *Imagery → Process → Difference*.
- Create an NDVI change composite: select your two NDVI layers (Control+click) in the *Contents*, then select *Imagery → Process → Composite*. Then, under the *Symbology* menu for your Composite image, change from "Stretch" to "RGB." You can then select which bands to display in each color. Play around with the combinations (this can take some time on the lab computers, as the bands don't come through properly labeled) so that you can get a clear picture of change, like this:



Q6: What differences do you see between 1987 and 1988 NDVI, and why? Include a screen capture of any of the images you've created that helps you to answer this question.

Q7: What do the red areas in this composite display represent (at least, red the way I did it, which displayed 1987 in Red and 1988 in Blue and Green; you might have a different combination)? If you had to guess, which direction do you think the winds were blowing during the 1988 fires? [This may be clearer in the Map_3D view; you can convert your current map to 3D view using* Convert *under the* View *menu.]

- Using the *Explore* tool, scan around to get a feel for typical NDVI values in the affected areas in 1987 and 1988.

- Under *Imagery → Raster Functions*, select Binary Thresholding and apply it to your NDVI Difference layer. This tool will divide pixels into two classes based on a threshold derived from image statistics.
- Right click on your output image and *Create Chart → Histogram*. You can hover over bars on the histogram to get a count of the number of pixels in each class.

Q8: Based on your NDVI difference threshold, what was the total burned area in 1988?

Q9: Name at least two potential sources of error in your calculation.

Unsupervised Classification

The NDVI approach to estimating burned area is an example of a differencing technique: we took a pre-burn image as our baseline, assumed that no major changes other than the burn had taken place (or took steps to account for any other changes), and defined burned area on the basis of change between two images.

Another approach is direct classification: if a burned forest has its own distinctive spectral signature, then we should be able to classify burned area on the basis of our 1988 image alone. One way to do this is through supervised classification. Since we have a reasonably good idea of where the burned areas are, a supervised classification would be a feasible and, probably, a preferable approach in this application. But we did that already in this course and it was time consuming, so this week we'll save the effort and just do an Unsupervised Classification.

- Following the same procedure as we used in Exercise 4, perform an ISO Cluster unsupervised classification
- Choose whatever maximum number of classes you think will be optimal. You can try several times with different maximum number of classes.

Q10: List the land cover "classes" that you've identified in the image. Include a screen capture of your classification in your answer.

Q11: Identify one or more classes that correspond to the burned areas in the image, and record the classified burned area.[1]

Returning to the *Spectral Signature* tool, examine the signature of burned areas relative to other land cover types that you are able to name in your classification.

Q12: What are the defining features (if any) that distinguish your burned areas class from other land cover classes in the scene?

That's all for Exercise 7. Please remember to transfer any files off of the lab computers that you want to keep to an external drive or web storage space. The final two questions for your report are:

Q13: Which result are you happier with, the NDVI threshold area estimate or the Unsupervised Classification estimate? Why do you think that one technique worked better than the other?

Q14: List at least two other techniques or combination of techniques that you could have used to estimate burned area.

[1] To calculate area, type "Zonal Statistics as Table" into the search bar in the *Analysis→Tools* geoprocessing dialog. Enter your classified image as both input datasets and "Value" as the zone field. The table will appear at the bottom of the *Contents* window, and you can right click on it to open it.